

claude-windows / 8d4866bf-8edd-4da9-909e-022ddee12675

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\8d4866bf-8edd-4da9-909e-022ddee12675\subagents\workflows\wf_4e32c8b8-725\agent-a5c2569e341c71ed9.jsonl`  
- SHA-256: `87fc196cb3bd9a47c8e6d66d5608f44962803476469d0486922eef84e0c96ffe`  
- Source modified: `2026-06-19T17:19:51+00:00`  
- Imported at: `2026-07-05T16:48:25+00:00`  
- project: `wf_4e32c8b8-725`  
- session_id: `8d4866bf-8edd-4da9-909e-022ddee12675`
```

Transcript

1. user (2026-06-19T17:18:10.919Z)

You are auditing the docs/ folder of the badminton-highlight-indexer repo to decide which docs can move to docs/archives/. Be rigorous and EVIDENCE-BASED ? the owner has run two doc-staleness sweeps and cares deeply about not archiving live work or breaking links.

REPO ARCHIVE POLICY (docs/archives/README.md) ? APPLY STRICTLY:

- Archive ONLY work in a clear TERMINAL state: DONE / COMPLETED / REJECTED / fully SUPERSEDED. "Never archive in-flight or pending work."
- KEEP-LIVE (do NOT archive) anything that is: living/active, in-progress, a not-yet-started but still-intended design/plan (owner-gated future work is LIVE, not terminal), a reference/navigation doc (CODE_MAP, README, HOW_RALLY_DETECTION_WORKS, EVALUATION), a template (PROJECT_DOC_TEMPLATE), or a roadmap/backlog/strategy doc still being tracked.
- Destinations inside archives/: past_projects/ (bounded completed project / tech-debt / starter effort) · research/ (completed or superseded research notes/plans) · decisions/ (ADRs) · checkpoints/ (dated state snapshots) · archives/ top-level (one-off audits/reports/launch records). Naming convention for a completed doc: <NAME>-COMPLETED-<date>.md.
- ALREADY-ARCHIVED COPIES EXIST for these (so the top-level file is likely a pointer-stub or a stale DUPLICATE to reduce, NOT a fresh move): docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md, docs/archives/DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md, docs/archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md. If your doc already has an archived copy, check whether the top-level is a stub (leave it) or a full duplicate (flag DEDUPE).
- CONTRADICTION TRAP: a doc may say "COMPLETED" yet still be cited as a LIVE source elsewhere (e.g. CLAUDE.md cites DEFERRED_HARDENING_PLAN.md #16 as the next platform slice). Before recommending ARCHIVE, grep CLAUDE.md, docs/NEXT_STEPS.md, docs/README.md, and other top-level docs for the filename. If it is referenced as a live/next source, verdict = ASK_OWNER (note the contradiction), not ARCHIVE_NOW.

YOUR BATCH (read each from docs/<name>): RALLY_QUALITY_RESEARCH.md, RALLY_DETECTION_GAP_CLOSURE.md, RALLY_DETECTION_QUALITY_REPORT.md, PROMINENT_COURT_DETECTION.md, QUALITY_ITERATIONS.md, R2_EVAL_METRIC_DESIGN.md, MULTI_SIGNAL_FUSION_PLAN.md

For EACH doc:

1. Read the status header + TL;DR/intro + (if present) the \$7/\$8 progress tracker + definition-of-done. You do NOT need to read every line of long docs ? sample enough to judge lifecycle state honestly.
2. Decide lifecycle and verdict per the strict policy. Remember: owner-gated / not-yet-started DESIGN docs are LIVE future work (KEEP_LIVE), NOT terminal. Only DONE/COMPLETED/REJECTED/fully-superseded is terminal.
3. Check for an already-archived copy (the policy lists three; also Glob docs/archives for the doc's stem). If the top-level is a thin pointer-stub ? KEEP_LIVE (it's intentional). If it's a FULL DUPLICATE of an archived COMPLETED copy ? DEDUPE_STUB (recommend reducing top-level to a stub or removing).
4. Cross-check "cited as live": Grep CLAUDE.md, docs/NEXT_STEPS.md, docs/README.md, and the other top-level docs/ for this filename. If it's cited as a live/next/authoritative source, it

CANNOT be archived even if it says COMPLETED ? ASK_OWNER with the contradiction noted.

5. Inbound references: Grep the whole repo (docs/, backend/, CLAUDE.md, *.py) for the filename to count link-breakage risk; list up to 6 key referrers.

Return the structured object with one entry per doc in your batch. Be decisive: most reference/template/active/design docs are KEEP_LIVE; reserve ARCHIVE_NOW for genuinely terminal single-efforts/reports/completed projects, and ASK_OWNER for the contradictory/ambiguous ones.

2. assistant (2026-06-19T17:18:13.463Z)

I'll audit each doc in my batch. Let me start by reading the archive policy and the key cross-reference files, then read each doc.

3. user (2026-06-19T17:18:14.446Z)

```
1      # archives/
2
3      Umbrella directory for all **historical and completed** material.
4
5      ## Live vs Historical
6      - **Live docs** (for ramp-up, constant updates, and current work) live at the top level
of `docs/`.
7      - Everything here is no longer the primary source of truth. It is kept for context,
decision history, and regression detection.
8
9      ## Classification inside archives/
10     - `past_projects/` ? Completed small tech debt, starter projects, or low-regret
initiatives (e.g. the original low-regret moves + scaffolding work). These are "past" once they
reach a terminal state.
11     - `decisions/` ? Historical ADRs and major architectural decisions.
12     - `research/` ? Completed or superseded research notes and plans.
13     - `checkpoints/` ? Snapshots of major project states (especially useful for detecting
regressions in quality, priorities, or baselines over time as we do quality iterations).
14     - Other one-off audits, etc.
15
16     ## Archival Rule
17     Only move something into `archives/` (or its subdirectories) when the work is in a clear
**terminal state**:
18     - DONE
19     - COMPLETED
20     - REJECTED
21
22     Never archive in-flight or pending work.
23
24     When archiving a project/phase:
25     1. Ensure the main document has an explicit **STATUS: COMPLETED** (or DONE/REJECTED)
header at the top.
26     2. Link back to the current living documents in `../` (e.g. `COMMERCIALIZATION.md`,
`BACKLOG.md`).
27     3. For regression detection value, consider adding a dated entry under `checkpoints/`.
28
29     See `past_projects/README.md` for the specific guidance on that category.
30
```

4. user (2026-06-19T17:18:15.475Z)

```
1      # Next steps (live) ? owned-model + multisport
2
3      **Updated:** 2026-06-13 (**rally-detection quality track**: fusion + experiment-harness
design docs merged,
4      owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior:
2026-06-10 (audit?M0?Gemini-battery session; later same day the **UI-alpha track opened**:
5      ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-
first delivery ? see
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6      `docs/UI_ALPHA_EXECUTION_PLAN.md`; **PR-1 async jobs (#16) MERGED 2026-06-11, PR #87**;
```

6

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7      upload/download MERGED 2026-06-11, PR #99** ? live-E2E-verified (3-part chunked upload
MD5 byte-identity +
8      compile?browser-download); next: PR-3 identity. Same day: the **8?9 quality sweep
#90?#98 landed** ? ruff/mypy/
9      coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin
collapse into
10     `trajectory_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626
re-verified exact).
11     **Authoritative roadmap:**
12     `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ ADR-008 for topology). **Latest blow-by-
blow:** memory
13     `current-state.md` / `session-handoff.md`. This file = the prioritized "what to do
next," refreshed each session.
14
15     ## ? Next-session / GPU-box pickup (2026-06-15)
16     **M0 in-container quick wins are COMPLETE:** ship-gate provisional blocker (**#170**),
train/eval split guard (**#171**), the **doc-consistency sweep** (**#174** ? corrected 16
drifted docs vs the code), and the **P3 person-feature fusion litmus machinery is already
present + suite-green** (`backend/eval/fusion_compare.py` + `ablation.py`: person-driven ?F1 /
bootstrap CI / paired sign test; the real-golden `skipif` litmus waits only on GPU-extracted
data). ? CI is red repo-wide due to a **GitHub Actions minutes/billing outage** (account-level,
not code) ? work is local-gated + `--admin`-merged until Actions is restored (Settings ? Billing
? Actions minutes / spending limit).
17
18     **Immediate next step ? the real P3 LOVO measurement** (this box has a GPU +
`torch+cu124` + `cv2`; videos and trajectory CSVs are present, but the GPU-extracted golden
features + `rallies.csv` labels are not yet here):
19     1. **[owner / GPU box]** `python -m backend.eval.fusion_golden --manifest
output/human_lovo_manifest.json --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-
data\features\2026-06-15-n6" ? produces `*_golden_features.json` for the 6 golden videos. (GPU
+ WSL + videos + trajectory CSVs + `rallies.csv` labels are owner-side only.)
20     2. Artifacts land in the **shared cloud folder** `C:\Users\avidu\OneDrive\rally-
annotator\golden-data\` (OneDrive auto-syncs both ways). Convention + workflow:
**[`GOLDEN_DATA_SHARING.md`](GOLDEN_DATA_SHARING.md)**; tracked in **#175**.
21     3. **[CPU dev/agent session]** `python -m backend.eval.fusion_compare --features-dir
<shared>\features\2026-06-15-n6 --record` + `python -m backend.eval.ablation --features-dir
<shared>\... --granularity group` ? real person-feature ?F1; copy the JSONs into `output/` to
also pass the `skipif` litmus tests (`tests/test_ablation.py::test_litmus_reproduces_gate_a`).
22
23     **Also queued (owner-gated / pending input):**
24     - Ship-gate target calibration ? **#172** (owner decision; needs corpus growth).
25     - Tests reorganization ? blocked on the owner's grouping metric (provide it ? an agent
will do it).
26     - Restore CI ? the Actions outage is account-level (billing/minutes), not a repo/config
bug.
27
28     **Leaving (bulky / M2 ? no start without owner go):** ActionFormer (M0.4), event-index
DB migration (M0.1), Gemini-silver purge (M0.3), cost-to-Gen-2 benchmark + ByteTrack
deprecation, multisport, repo rename, PMF execution.
29
30     ## Where we are (one paragraph)
31     ADR-008 ratified ("repo split driven by productionization, not isolation"): training
lives in-repo (`training/`)
32     behind a CI-enforced import wall; the featurizer is single-sourced
(`backend/features/rally_seq.py`, train==serve);
33     the **Gen-0 harness ships** (`python -m training.gen0.harness --manifest
output/human_lovo_manifest.json --floor
34     eval_baselines/heuristic_lovo_n6.json --version <semver>`) and produced artifact v0.1.0
(LOVO **0.626**, floor passed,
35     sign test 5/6 wins p=0.109 ? honestly not significant, ship=False). The **n=6 owned
corpus is in the collector**
36     (262 rallies, Proprietary, commercial export = owned data only after the ShuttleSet
reclassification). The **Gemini
```

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37     battery is done** (see table) ? the moat is quantified.
38
39     ## The quality table that now drives strategy (IoU 0.5 vs human golden labels)
40     | Path | Mahadevpura (clean-ish) | Kushagra (multi-court) |
41     |---|---|---|
42     | Heuristic (velocity windowing) | 0.657 | 0.157 |
43     | **Gen-0 owned head** (LOVO, n=6) | 0.744 | 0.561 |
44     | Two-stage WASB+Gemini refine | 0.83 | ? |
45     | **Gemini wrapper** (`gemini-flash-latest`) | **0.93** | **0.66** |
46
47     **Reading:** clean footage is commoditized (serve it with Gemini for cents). **Multi-
48     court drops the commodity
49     wrapper to 0.66 ? that is the owned model's ship target** (its FPs are background-game
50     pickups + dead-time merges,
51     the exact contrast-metric confound). Gen-0 is 0.10 behind with only n=6 ? a corpus-
52     growth-sized gap, not an
53     architecture-sized one. Two-stage refine = the cost-efficient serving path for LONG
54     tapes (uploads candidate clips
55     only); wrapper 503-flakiness (observed all session) makes the provider-fallback registry
56     load-bearing.
57     Baselines tracked in `eval_baselines/` (heuristic_lovo_n6.json,
58     gemini_wrapper_2026-06-10.json).
59
60     ## Rally-detection quality track (design merged 2026-06-13, owner-gated; implementation
61     in progress)
62     **This is the track a desktop/local agent is slated to pick up.** Design docs landed (PR
63     #112 + session). All implementation is **owner-gated** (per explicit in the plan); each step =
64     ONE PR off `master`.
65
66     **Hardening loop (T1-T5 positive tests + audit + finale) COMPLETE (2026-06-14, PRs
67     #156/#157/#160/#161/#163 + #165 + final records).** See archived
68     `docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` (Living Log with every
69     pre/post/review) and `docs/archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md` (full rules
70     + STATE). Top-level now pointer stubs per #165 review. All 5 items + verification + archive
71     done; clean slate.
72
73     - **Docs:** `docs/MULTI_SIGNAL_FUSION_PLAN.md` (shuttle + person + audio, one fusion
74     layer) .
75     `docs/EXPERIMENT_HARNESS.md` (A/B + shadow + promotion gate, zero-regression) .
76     `docs/ROORA_LABELING_GUIDELINES.md` (v8) . `docs/HOW_RALLY_DETECTION_WORKS.md`
77     (2-layer mental model).
78     - **Key finding (no re-investigation needed):** the owner's "held-shuttle counted as a
79     rally" bug is **NOT a missing capability** ? `backend/pipeline/detectors/rally_gate.py` already
80     implements the net-crossing **"Gate A"** (`apply_gate(..., min_crossings=2)`) that kills
81     service-feed/held-shuttle windows, but it is **only called from eval drivers** (and there was no
82     `min_crossings` knob in `IndexingConfig`). **Wiring Gate A into the live segmenter (now via
83     fusion_hybrid) was the single cheapest, highest-confidence quality win.** (P2 FusionSegmenter
84     specifics: registry `fusion_hybrid`, **default-OFF**, seeds from substrate, persists
85     `net_crossings` + optional person features, optional served Gate A/B via config knobs;
86     adversarially reviewed; suite green. See `tests/test_served_gate_a_regression.py` for the honest
87     finding that on production windowing Gate A is neutral/negative in some cases, hence kept opt-in
88     with knob for safety, and the leaderboard revert note.)
89     - **Current status (2026-06-14):** P1 (person cue extraction to reusable
90     `PersonDetector`) + P2 (FusionSegmenter + Gate A signal + served handoff gating) + experiment
91     harness P1 DONE (adversarial review + NO-REGRESSION; suite green). **P3 next (this PR + follow-
92     ups):** learned fusion (person features ? harness `compare` LOVO lift on the 6 golden videos;
93     eager-person-compute optimisation); then P4 audio via hit_detector.
94     - **Pick-up order (each = ONE PR off `master`, owner approval first, flag-gated default-
95     OFF, promote only on measured LOVO):**
96     1. (P2 first-step, largely done via fusion_hybrid) Wire Gate A into production served
97     path + config knob (min_crossings); apply in hybrids at runtime. (Now available opt-in; served
98     neutral on production windowing per litmus ? kept opt-in with knob for safety.)
99     2. P3 first step (this work item): harness `compare` LOVO litmus for person-on variant
100    (players_in_court + two_sided + colocation etc. from `fusion_features`) on the 6 golden; add the
101    eager compute optimisation note/landing. Pure + harness (testable here).

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67     3. P3/P4 continuation: audio cue integration + full learned promotion if lift proven.
68     - **Discipline (from EXPERIMENT_HARNESS.md):** every new cue **flag-gated, default
69 OFF**; promote to default **only on measured LOVO lift** through `run_regression.py` (exit
70 0/1/3); pinned `CONTROL` ? rollback = config flip. Harness re-uses ~80% existing code.
71     - ? Gate A, person cue logic, harness, and pure fusion paths are testable here; the
72 heavier real-video / golden LOVO / ablation confirmation waits on the GPU-extracted golden
73 features + `rallies.csv` labels, which are not yet present in this checkout.
74     - **External due-diligence review inputs (2026-06-17) ? PICK UP AFTER current ideas are
75 exhausted; full details + citations in [`RALLY_QUALITY_RESEARCH.md`](RALLY_QUALITY_RESEARCH.md)
76 §5.6.** An owner-commissioned 31-reference review of the quality report *validated* the
77 methodology and surfaced these **future** levers (research inputs, not active work, not new
78 claims; verify citations before any submission):
79     1. **"First-mile" paper framing (highest-value):** stroke-classifiers (BST/DSTA) + the
80 big datasets (ShuttleSet ~36k strokes, Fine-Badminton ~27k actions) all **assume a pre-trimmed
81 rally clip already exists** ? we solve the ignored prerequisite. Pitch the paper as **the
82 missing untrimmed-video preprocessing pipeline that lets BST/MonoTrack run on raw footage."**
83     2. **Ground R2 in Action-Spotting / Precise-Event-Spotting**
84 (SoccerNet/TennisSet/T-DEED) ? turns R2 into a defensible publishable contribution.
85     3. **Add domain baselines for any submission:** MonoTrack + BST on ShuttleSet / Fine-
86 Badminton (we have only heuristic + Gemini today).
87     4. **Prefer TemporalMaxer over ActionFormer** for the M0.4 scorer swap (parameter-free
88 max-pool, ~2.8× fewer GMACs, small-N robust) ? see "M0.4 next increment" below.
89     5. **Court polygons ? pose ? the rally-END lever:** masking on-court players can
90 resolve *occluded end-of-rally states* (the #1 remaining gap vs Gemini), not just spectator
91 noise.
92     6. **Name the eval?serve lesson "pipeline distribution skew"** ? a citable paper
93 "lesson."
94
95 **Discipline reminder:** owner approval before starting any owner-gated item; one PR per
96 step; babysit the PR (poll/fix/reply) until merge observed, *then* advance.
97
98 ## Prioritized next steps
99     1. **[owner, in progress] GROW THE GOLDEN CORPUS** ? the single highest-leverage move;
100 every lever below feeds on it.
101     Priority order for new videos: **a) multi-court badminton** (the target distribution
102 AND where the ship target
103 lives), **b) ?3 matches per new sport ? table tennis first** (WASB transfers at F1
104 0.909; pickleball labels are
105 corpus-gold but not trainable until a clean MIT/Apache ball detector exists), (c)
106 capture variety per
107 `VIDEO_RECORDING_GUIDELINES.md`; **adult sessions only for the training pool**
108 (consent guardrail). Per video:
109 label ? `curate import-local --normalize --license Proprietary --fps <true fps>` ?
110 re-run the Gen-0 harness.
111     The paired sign test needs n: at n=10 with 9 wins, p?0.011 ? significance is
112 reachable this month.
113     2. **[owner decision] Set the ship-gate target** ? now informed: floor = heuristic 0.480
114 (necessary), **multi-court
115 reference = wrapper 0.66** (the number that makes the owned model worth serving),
116 clean-footage ceiling = 0.93
117     (not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@IoU0.5 ? 0.70
118 on multi-court folds with
119 paired-sign p<0.05 vs heuristic" ? owner to ratify/adjust.
120     3. **M0.4 next increment ($0/local):** swap the Gen-0 LogReg for a lightweight temporal
121 head (MIT, minutes on the
122 2070S ? NOT the heavyweight Gen-2 stack) inside the existing harness; per-video
123 threshold calibration (the
124 LOVO-vs-oracle gap is visible per fold in the harness output). **Prefer
125 TemporalMaxer** (parameter-free
126 max-pool, ~2.8× fewer GMACs, small-N robust) over ActionFormer/ASFormer per the
127 2026-06-17 external review
128 (`RALLY_QUALITY_RESEARCH.md` §5.6) ? better fit for the cheap/local/fast mandate.
129     4. **Multisport thread:** ingest Roora pickleball/TT CSVs (`import-local --normalize`);
130 merged-prototype LOVO across
131 badminton+TT once TT has ?3 matches; identify a clean pickleball ball detector. ? TT-

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on-WASB serving stays gated
99 by C3 (provenance multiplies per sport).
100 5. ****PMF validation (cheaper than any engineering month):**** C13 academy interviews (+
the two added questions:
101 "multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot
at the warmest 2-3 academies
102 (consent at capture; demo served via the Gemini path or human-polish ? the reel
pattern exists:
103 `output/MahadevpuraSingles\_highlights.mp4`). ****Field kit COMPLETE (2026-06-12, owner
ratified the
104 physical-visit approach; reframed product-feedback-first per owner spec):****
105 `docs/PRODUCT\_FIELD\_ASSOCIATE\_FLYER.md` (the recruiting flyer ? owner fills ?/contact
placeholders and recruits) +
106 `docs/FIELD\_VISIT\_PLAYBOOK.md` (the hired associate's visit playbook: coach + student
script, demo-after-Q6
107 rule, ground rules, Phase-2 record-and-show-back stretch goal) +
`docs/FIELD\_VISIT\_ANSWER\_SHEET.md` (per-visit
108 capture form incl. demo-reaction + student blocks) + `docs/PMF\_FIELD\_SIGNALS.md`
(G/A/R anchors + ****pre-registered
109 decision rules**** ? PROCEED / PIVOT A-C / FALSIFIED thresholds fixed before the first
visit).
110 6. ****Housekeeping:**** ~~rotate the Gemini API key~~ ****DONE 2026-06-10**** (owner-verified
working; confirm old-key
111 deletion in AI Studio); ****rename repo off "badminton"**** (owner action,
112 in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the
battery completed.
113
114 **## Pointers / artifacts**
115 - Roadmap: `docs/OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md` · Topology: ADR-008
(`docs/archives/decisions/DECISIONS.md`) ·
116 Study: `docs/archives/research/OWNED\_MODEL\_TRAINING\_STUDY.md` · Quality history:
117 `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
118 - Harness: `training/gen0/harness.py` (one command ? train ? LOVO ? gate ? artifact
bundle; weights gitignored,
119 manifest reviewable). Featurizer: `backend/features/rally\_seq.py` (FEATURE_SCHEMA v1).
120 - Corpus: collector DB (system-of-record, 6 matches/262 rallies) · eval manifest
`output/human\_lovo\_manifest.json`
121 (gitignored) · durable trajectories + Roora files: `C:\Users\avidu\Projects\Annotation
Setup\Collect\Trajectories\` ·
122 labels: `...\Collect\Golden Labelled\`.
123 - Gemini ops learnings: serial uploads (`indexing.ai\_max\_workers=1`) + chunk re-encode
124 (`indexing.ai\_chunk\_scale\_width=1280`) + 6-attempt exponential generate retry ? a
cold/prepaid project burst-throttles
125 with a MISLEADING "credits depleted" 429; ramp gradually.
126 - ? No owned-model implementation beyond greenlit M0.x until owner greenlights the next
milestone. Committed next
127 PLATFORM slice = async job model (single-tenant).
128

5. user (2026-06-19T17:18:15.509Z)

1 - ROORA_LABELING_GUIDELINES.md ? Latest (v8) vendor
labeling guidelines mirrored from the collector's `docs/annotation/`, with the fusion-driven
court-calibration / singles-doubles additions marked `? ADDED (fusion)`.
2 - RALLY_QUALITY_RESEARCH.md ? ****Foundation doc**** for
closing the local-vs-Gemini rally-detection gap: the measured gap + the code-grounded over-merge
diagnosis ("shuttle moving ? rally"), an inventory of the eval/experiment harness + its gaps,
the strengthened evaluation framework (segmental Edit/F1@k that make over-segmentation first-
class, golden-corpus + active learning, Gemini-agreement as a *shadow* signal), an
****adversarially-verified, cited external-research synthesis**** (TAL/TAS boundary heads, racket-
sports rally-state cues, small-data, eval rigor), a ****prioritized cheap?durable roadmap with
falsifiable success criteria****, and (\$7) a ****tracked project plan**** ? work-item checklist,
sequencing, the single quality ****number**** attributed to the effort, and a ****definition of done****
(project completes when all tiers land + the number is recorded). The plan that the
QUALITY_ITERATIONS.md log executes against; ties together

EXPERIMENT_HARNESS / MULTI_SIGNAL_FUSION_PLAN / ANALYZERS_AND_RECONCILERS /
OWNED_MODEL_IMPLEMENTATION_PLAN.

3 - RALLY_DETECTION_GAP_CLOSURE.md ? **Tracked execution doc (2026-06-18)** for closing the remaining rally-detection accuracy gap **per video, before launch**, driven by the growing golden corpus. First ground-truth clip (GX010141) reframed it: boundaries are tight (R2/R3/R4 holds) ? the gap is **detection**: (G1) recall (local & Gemini both miss ~half on hard clips, and *different* rallies ? complementary), (G2) precision (local over-fires), (G3) **rally-END over-extension on the quick-return-after-point** (shuttle still moving ? needs a *player-state* signal). Levers (owner-gated): player-kinematics + shuttle-in-hand / sustained-invisibility end-rule with reverse-timeline backtrack, `rally\_gate` Gate A wiring, complementary fusion. **Peer** of `RALLY\_QUALITY\_RESEARCH.md` (execution tracker, not research); §7 is the source of truth.

4 - PROMINENT_COURT_DETECTION.md ? **Design + experiment (2026-06-19, default-OFF, owner-gated flip)** for the **multicourt G2 over-fire**: identify the **prominent (foreground) court** (the one covering the majority of the frame) and keep only rallies whose shuttle lives inside it ? re-defining the shuttle signal `(x,y,visible)` ? `(x,y,visible-AND-inside-prominent-court)` *before* windowing, so a background/adjacent-court shuttle never forms a rally. 2D floor polygon (MVP) ? **pseudo-3D play-volume** (refinement, so high clears aren't clipped). Born from the GX010142 rally-#1 background-court FP; reuses `fusion\_features.point\_in\_polygon` + `FusionConfig.court\_polygon`. §8 progress tracker is the source of truth.

5 - RALLY_DETECTION_QUALITY_REPORT.md ? **Technical-paper checkpoint (2026-06-17)** of rally-detection quality: the honest measured state (over-segmentation milestone tolerance-F1 0.09170.323 at n=11, p=0.002; merge-rate 0.69470.150; first real-footage ground-truth at-par with START at Gemini parity), the task-faithful R2 metric, a detailed Gemini comparison, next steps + expected wins, areas yet to be explored, and a publishability / path-to-submission verdict. Renders to a polished PDF; the externally-shareable "toward a paper" summary of `RALLY\_QUALITY\_RESEARCH.md` + `QUALITY\_ITERATIONS.md`.

6 - EVALUATION.md ? How we measure and improve quality.

7 - EXPERIMENT_HARNESS.md ? Design (P1 SHIPPED) for A/B + shadow + SxS experimentation on rally-detection variants with **zero production regression** ? experiments are decoupled from deploy (new cues default OFF, promote only on measured LOVO lift, rollback = flip a flag). Composes the existing `backend/eval/` machinery (`calibration`, `regression`, `metrics`); implemented in `backend/eval/experiment.py`.

8 - QUALITY_ITERATIONS.md ? **Living log** of rally-detection quality iterations (hypothesis ? what shipped ? measured/expected effect ? lesson). The reverse-chronological record; terminal-state snapshots get archived under `archives/quality-iterations/`.

9 - ANALYZERS_AND_RECONCILERS.md ? Design (owner-gated, NOT started): a **signal-fusion framework** for rally detection ? producers emit confidence-scored **signals** at three temporal **scopes** (frame detectors / window analyzers / session analyzers), a learned **scorer** weights them, and an auditable **report-first reconciliation** pass lints the decision against the evidence. The spine: **no special-casing in the decision path** (signals?scorer, never `if/else`). Worked examples: a service-fault analyzer, a warm-up/practice span detector, and consuming a per-frame shuttle-state detector. A peer to `EXPERIMENT\_HARNESS.md` and `MULTI\_SIGNAL\_FUSION\_PLAN.md` (ADR-007 modular rally layer).

10 - PERSONALIZATION_PLAN.md ? Design (owner-adopted **hybrid**;
Phase-0 landing) for the **per-user personalization feature on a generalization-first baseline**: a thin per-user *tuning* layer gated by the honest small-N statistics (a **min-N promotion floor** + a **structural-guard exemption**, never a service gate) + a **contribution flywheel** (free-tier videos pool into the owner-trained baseline; *a tool that gets better the more you help it*). Records the **locked owner decisions** (identity, storage, `min\_n` semantics, Gate-A default, free-tier pooling). Built so far: per-user promotion gate (#141) + held-out-USER learning curve (#142).

11
12 - GOLDEN_REGRESSION_FIXTURES.md ? **Design (owner-gated, NOT started)**: video-free, non-attributable regression fixtures so a clean clone runs the rally-seg suite with **no video/GPU/DB**. Two layers ? the post-detector freeze + **dual gates** (characterization + quality-floor) reusing the `backend/eval/` stack, and the **SEALED-FIXTURES** privacy filter (de-attribution-at-source, tiered public/key-gated, shuttle-only public tier). Carries a cited **IN/US/EU legal memo** (derived-data IP/privacy), a threat model, and a **§7 progress tracker + definition-of-done**. Extends
GOLDEN_DATA_SHARING.md.

13

14 ## Product / platform plans

```

15      - [DESKTOP_APP_PLAN.md](DESKTOP_APP_PLAN.md) ? **Tracked scoping doc (2026-06-18, owner-
gated, `[design]` ? NOTHING built):** package the local pipeline as a **cross-platform desktop
app** for non-technical players ? plug in the camera's USB/SD card, click once, get back **only
the highlight reels**, fully on-device (no upload, no original-video bloat). The centerpiece is
the **de-WSL native-compute port** (run WASB natively per-OS: Linux/Windows CUDA · macOS MPS/CPU
? drop WSL2), and its **make-or-break unknown**: is CPU acceptable on a typical *no-GPU*
enthusiast laptop (WASB is ~3.4× real-time on a laptop 2070S)? $7 tracker = **P0 compute-
feasibility spike ? P1 native detector ? P2 app shell ? P3 packaging/installers**. Realizes the
**L0 fully-local** tier of [VIDEO_LOCALITY_MODEL.md](VIDEO_LOCALITY_MODEL.md); is the
**`LocalDesktop` ComputeBackend** of [PLATFORM_ARCHITECTURE.md](PLATFORM_ARCHITECTURE.md) §3.2;
the smaller/export-clean owned model
([OWNED_MODEL_IMPLEMENTATION_PLAN.md](OWNED_MODEL_IMPLEMENTATION_PLAN.md)) is one answer to the
compute crux.
16
17      ## Practical / ongoing
18      - [TRACKNET_WSL_SETUP.md](TRACKNET_WSL_SETUP.md)
19      - Golden set docs (`GOLDEN_SET`)
20      - Evaluation harness and calibration drivers (in `backend/eval/`)
21      - [GOLDEN_DATA_SHARING.md](GOLDEN_DATA_SHARING.md) ? How the GPU box and CPU dev/agent
sessions share golden artifacts (trajectory CSVs, `*_golden_features.json`) via a OneDrive
folder; the GPU-extract ? CPU-litmus split. See **#175**.
22
23      ## Conventions & templates
24      - [PROJECT_DOC_TEMPLATE.md](PROJECT_DOC_TEMPLATE.md) ? Reusable skeleton for a **tracked
project doc**: status header · decisions-locked · progress tracker (??/?) · definition-of-done
· archived on completion. Standardizes the emergent pattern in `RALLY_QUALITY_RESEARCH.md` §7
and `GOLDEN_REGRESSION_FIXTURES.md`. Use for any substantial design/project work.
25
26      ## Historical documents
27      Completed plans, old research, ADRs, and checkpoints live in [`archives/`](archives/).
28
29      - [CODE_AUDIT_AND_TEST_HARDENING.md](CODE_AUDIT_AND_TEST_HARDENING.md) ? **COMPLETED**
(2026-06-14; T1?T5 + full loop + verification per #162). Living record of the June 2026 plumbing
/ contract / architectural audit, 626-test infrastructure, positive test plans for the 5
identified issues, multi-layer regression protocol (including ablation/experiment layer), and
pre/post-change verification checklist. See `docs/archives/CODE_AUDIT_AND_TEST_HARDENING-
COMPLETED-2026-06-14.md` and `docs/archives/checkpoints/2026-06-14-hardening-loop-complete.md`
for the final archived state. (Top-level is now pointer stub per review; full content in
archives/.)
30
31      See `archives/README.md` for the umbrella policy and `archives/past_projects/README.md`
for classification of small tech debt / starter projects.

```

6. assistant (2026-06-19T17:18:17.597Z)

Now let me read each doc in my batch. Let me start with the first few.

7. user (2026-06-19T17:18:18.568Z)

```

1      # Rally-Detection Quality: Research, Eval Framework & Project Plan
2
3      **Status:** Foundation doc (2026-06-16), run as a **tracked project** ($7) ? **"Bridge
the
4      local?Gemini rally-detection gap,"** **completed when all committed tiers land and a
single quality
5      number is attributed ($7.1).** The empirical + research backbone for closing the gap
between the
6      **local, no-cloud rally segmenter** and the **Gemini** path. Iterations land as their
own measured
7      PRs, logged in [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md); this doc is the plan
they execute against.
8
9      **How to read this:** $1 frames the gap with measured numbers. $2 is the precise, code-
grounded
10     diagnosis (read before proposing fixes). $3 inventories the eval/experiment harness we

```


already

11 have. §4 specifies how we strengthen that harness (the "really strong evaluation framework").

12 §5 is the cited external-research synthesis (4 dimensions). §6 is the prioritized roadmap with

13 falsifiable success criteria. **§7 is the project plan ? work-item tracker, sequencing, the single

14 number we attribute to the effort, and the definition of done.** §8 recaps the non-negotiable

15 constraints + cross-links.

16

17 > **This doc does NOT duplicate prior design.** It builds on, and cross-links:

18 > [`HOW\_RALLY\_DETECTION\_WORKS.md`](HOW_RALLY_DETECTION_WORKS.md) (the 2-layer mental model),

19 > [`EXPERIMENT\_HARNESS.md`](EXPERIMENT_HARNESS.md) (the A/B + LOVO harness, P1 shipped),

20 > [`MULTI\_SIGNAL\_FUSION\_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md) (shuttle+person+audio fusion),

21 > [`ANALYZERS\_AND\_RECONCILERS.md`](ANALYZERS_AND_RECONCILERS.md) (the signal-fusion framework +

22 > §13 literature review ? **already** establishes that our skeleton is standard TAL/TAS +

23 > late-fusion; we *cite, don't re-derive*), [`ABLATION\_LAYER.md`](ABLATION_LAYER.md),

24 > [`EVALUATION.md`](EVALUATION.md),

[`OWNED\_MODEL\_IMPLEMENTATION\_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md)

25 > (TAL-head-first ? VLM), and [`GOLDEN\_SET\_VENDORING.md`](GOLDEN_SET_VENDORING.md) (corpus growth).

26

27 ---

28

29 ## 1. The gap (measured)

30

31 Two paths produce rally segments today. **Gemini** (cloud VLM, watches the video semantically)

32 is strong; the **local** path (WASB shuttle trajectory ? velocity windowing, no cloud) lags

33 badly. Measured F1 vs golden labels ([`QUALITY\_ITERATIONS.md`](QUALITY_ITERATIONS.md)):

34

35 | Path | Mahadevpura (clean, single court) | Kushagra (multi-court) |

36 |---|---|---|

37 | Heuristic velocity windowing (local today) | 0.657 | **0.157** |

38 | Gen-0 owned head (LogReg on 5 trajectory features, LOVO n=6) | 0.744 | 0.561 |

39 | Two-stage WASB + Gemini refine | 0.83 | ? |

40 | **Gemini wrapper (gemini-flash)** | **0.93** | **0.66** |

41

42 The local heuristic is **footage-fragile**: fine on a clean single court, collapses on

43 multi-court / busy venues. The owned Gen-0 head already closes most of the multi-court gap

44 (0.561 vs 0.66) ? the residual is **corpus-sized, not architecture-sized**.

45

46 ### 1.1 The 2026-06-16 local-vs-Gemini divergence run (no golden labels)

47

48 A fully-local run (`wasb\_hybrid --skip-ai-handoff`) vs Gemini on 3 fresh matches

49 (Kushagra/Yash June-12; report: `?/June 12 Local-vs-Gemini/\_LOCAL\_vs\_GEMINI\_REPORT.md`,

50 generator `scratch/compare\_local\_vs\_gemini.py`). **Gemini is a strong reference here,

NOT

51 ground truth** (these clips are unlabeled) ? so this is an agreement/divergence study:

52

53 | | Gemini | Local (WASB no-AI) |

54 |---|---|---|

55 | rallies / windows (3 videos) | 59 | 12 |

56 | total "play" | 6:28 | **17:41** |

57 | mean window duration | **6.1 s** | **65 s** (10.6x) |

58 | true misses (gemini-only) | ? | **0** |

59 | merge groups (one local window ? ?2 Gemini rallies) | ? | 7 |

60

8. user (2026-06-19T17:18:19.332Z)

```
1      # Rally Detection ? Closing the Accuracy Gap (golden-eval-driven, per-video)
2
3      > **Status:** `DRAFT` ? implementation **owner-gated** . **Owner:** Avi . **Created:**
2026-06-18 . **Last updated:** 2026-06-18
4      > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
5      > **Tracking anchors:** §7 progress tracker is the source of truth; pointer in memory
`current-state`; add to `docs/README.md` + `docs/NEXT_STEPS.md` when work starts.
6      > **Relation to existing docs:** **peer-of**
[ `RALLY_QUALITY_RESEARCH.md` ] (RALLY_QUALITY_RESEARCH.md) (that is the research + eval-framework
doc; this is the *execution tracker* for the remaining accuracy gap). **Builds on**
[ `MULTI_SIGNAL_FUSION_PLAN.md` ] (MULTI_SIGNAL_FUSION_PLAN.md) (signal mechanics),
[ `QUALITY_ITERATIONS.md` ] (QUALITY_ITERATIONS.md) (the R-log / empirical backbone),
[ `GOLDEN_VIDEOS.md` ] (GOLDEN_VIDEOS.md) (the corpus). Does **not** duplicate them ? it points.
7      > **Honesty note:** each claim tagged `[verified]` (checked vs code/measured) /
`[design]` (proposed). The headline numbers below are real measurements, but `n` is small ? read
§6.
8
9      ---
10
11     ## 0. TL;DR
12     We are using the **growing golden corpus** (human rally labels) to find and close the
rally-detection accuracy gap **per video, before launch** ? and the first ground-truth clip
already reshaped the picture. The boundary work (R2/R3/R4) **holds** ? when local fires on a
real rally, its edges are tight ( $|?| < 1$  s). **The remaining gap is DETECTION, not boundaries:**
(G1) **recall** ? both local and Gemini miss ~half the rallies on hard clips; (G2) **precision**
? local over-fires (false positives), especially on continuous drilling; (G3) **rally-END over-
extension** ? when a player *quickly knocks the shuttle back after a point*, shuttle-velocity
alone can't tell "rally" from "casual reset," so the window runs long / merges. n=2 sharpens the
priority: **LOCAL = high recall / low precision, GEMINI = high precision / low recall**
(opposite, complementary ? on the normal clip local's recall *beats* Gemini's). So **the #1
lever is precision** (cut local's false positives via `rally_gate` Gate-A + colocation; the
player-state rally-END rule feeds this), with **signal fusion** (local's recall + Gemini-style
precision) as the durable win. **Caveat:** two golden clips so far (n=15 corpus,
GX010141+GX010137) ? directional, not conclusive.
13
14     ## 1. Problem & goal
15     The detector is good enough to ship a highlights product, but the golden eval shows
clip-dependent accuracy that we want to **close consistently on every video, not just on
average** ? catching the failure modes in the quality loop instead of after launch.
16
17     - **Concrete target:** on every corpus video, local is **at-par-or-better than Gemini vs
the golden labels**, with **no per-video regression** on either eval set (the dual-eval-set
discipline, [[eval-dual-set-regression]]).
18     - **"Good" =** the three gaps below closed enough that a milestone flip (default-ON)
clears the over-seg guard AND improves per-video F1 broadly, recorded in a Launch Report
([[launch-report-convention]]).
19     - **Read first to get current:**
[ `RALLY_QUALITY_RESEARCH.md` ] (RALLY_QUALITY_RESEARCH.md) §1?2 (the gap + diagnosis),
[ `MULTI_SIGNAL_FUSION_PLAN.md` ] (MULTI_SIGNAL_FUSION_PLAN.md) §4 (in-flight vs in-hand),
[ `HOW_RALLY_DETECTION_WORKS.md` ] (HOW_RALLY_DETECTION_WORKS.md) (the 2-layer model).
20
21     ## 2. Decisions locked
22     | # | Decision | Source / date | Implication |
23     |---|-----|-----|-----|
24     | D1 | The gap is **DETECTION (recall + precision + rally-END), not boundary precision**
| golden GX010141, 2026-06-18 `[verified]` | Don't build a generic boundary head; spend on
rally-state signals. |
25     | D2 | Measure **per-video AND aggregate; no regression on ANY video** | [[eval-dual-
set-regression]] / [[decision-rigor-norm]] | A win must broadly help, not trade one clip for
another. |
26     | D3 | **Golden labels = ground truth; Gemini = reference, not truth** | n=15 corpus |
Gemini under-detects too ? agreement ? accuracy. Report both, gate on golden. |
```

27 | D4 | New signals land ****default-OFF + flag-gated****, promoted only on measured lift |
 28 | D5 | ****Rally-END needs a PLAYER-state signal**** ? shuttle velocity can't separate a
 competitive rally from a casual post-point return | owner insight + R4 limitation, 2026-06-18
 29 | Add person kinematics / shuttle-in-hand as the rally-end discriminator. |

30 ## 3. Foundation ? the measured gaps (golden ground truth)
 31 Ground-truth-scored Boxhill clips (more land as the owner labels them). Milestone
 windowing = cap6 + R4 ([`QUALITY\_ITERATIONS.md`](QUALITY_ITERATIONS.md) R3/R4). Local & Gemini
 vs truth, n=2 so far: `[verified]`

32 | clip (golden) | system | found | tIoU@0.5 P/R/F1 | tolF1 | hits | false+ | missed |
 33 \[?start\] | \[?end\] |
 34 |---|---|---|---|---|---|---|---|---|
 35 | ****GX010141**** (29, hard/drilling) | LOCAL | 22 | 0.59/0.45/****0.51**** | 0.43 | 13 | 11 |
 36 | 0.78 | 0.68 |
 37 | | GEMINI | 16 | 0.94/0.52/****0.67**** | 0.53 | 15 | 4 | 14 | 0.31 | 0.46 |
 38 | ****GX010137**** (34, normal match) | LOCAL | 39 | 0.77/****0.88****/****0.82**** | 0.68 | 30 | 14
 39 | 9 | 0.39 | 0.44 |
 40 | | GEMINI | 27 | 1.00/0.79/****0.89**** | 0.89 | 27 | 0 | 7 | 0.50 | 0.52 |

40 - ****Corpus aggregate (n=13, milestone):**** tIoU F1@0.5 ****0.474**** [0.359, 0.579], F1@0.25
 0.633; R2 tolF1 0.359. GX010137 (0.822) is the corpus's ****best**** clip; GX010141 (0.510) is above
 the mean. `[verified]`

41 - ****Opposite failure modes ? the central finding (n=2).**** ****LOCAL** = high recall, low
 precision; ****GEMINI** = high precision, low recall. ****On the normal clip GX010137, local's **recall**
 0.88 BEATS Gemini's 0.79** (local finds **more** real rallies) but local fires 14 false positives;
 Gemini fires ****zero**** false positives but misses 7 rallies. They are ****complementary**** (on
 GX010141 they overlap on only ~3 rallies yet each hits ~13?15 of the golden set). ? ****fusion**
 (local's recall + Gemini-style precision filtering) is the clear win.**

42 - ****The gaps, re-prioritized by the data:****
 43 - ****G2 ? precision is LOCAL's #1 gap**** (over-fires on both clips: 11/22 and 14/39
 false positives). This is the cheapest + highest-leverage fix (Gate-A / colocation). Filter
 local's FPs on GX010137 and it lands ~0.88+ ? at-par with Gemini, since local already out-
 recalls it.

44 - ****G1 ? recall**** is clip-dependent: a local **strength** on normal play (? Gemini), but
 on hard continuous-drilling clips **both** miss ~half (GX010141 recall ~45?52%). The recall lever
 matters mainly for the hard clips.

45 - ****G3 ? rally-END over-extension**** (quick-return-after-point): shuttle keeps moving ?
 window runs long / merges; R4 (shuttle-only) can't catch it. Contributes to G2's false
 positives/merges.

46 - ****Local's gap to Gemini SHRINKS on normal play:**** hard clip ??0.16 (0.51 vs 0.67) ?
 normal clip ??0.07 (0.82 vs 0.89), and local **out-recalls** Gemini there. The deficit is
 precision, not detection.

47 - ****What is NOT the gap:**** boundary precision. When local hits a rally its edges are
 tight (|?start|/?end| ? ~0.8 s, on GX010137 even tighter than Gemini). The R2/R3/R4 windowing
 is doing its job. `[verified]`

48 - ****Long-rally lens refinement (P7, n=15, `[verified]`):**** filtering the eval to >5s
 rallies (the eye-catching highlights) shows local's single-court over-fire is ****short-FP-**
dominated** (seg_ratio 1.43?1.04 ? duration-filterable), but ****multicourt still over-fires 1.58x**
 on LONG rallies** (adjacent-court games are genuine long rallies). ? the multicourt precision
 gap needs ****court-localization (Gate-B / court mask, Lever C)****, NOT a duration filter; and the
 strict tolerance precision is gated by ****over-merge contamination**** of the long-pred set, not
 just short FPs. See [`QUALITY\_ITERATIONS.md`](QUALITY_ITERATIONS.md) "P7".

49 - ? ****Data caveat:**** GX010141 was labeled before ~3 rally-annotator-tool bugs were found
 + fixed (GX010137 is post-fix). GX010141's numbers may shift on a re-export ? re-verify before
 drawing hard conclusions from it.

50

51 ## 4. Design / architecture ? the levers (REUSE MAP)
 52 The signal mechanics live in
 53 [`MULTI\_SIGNAL\_FUSION\_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md); this is which lever attacks which
 gap.

54 - ****Lever A ? player-state rally-END rule**** (targets ****G3 + G2****) `[design]`. Generalize

R4 from shuttle-only to ****shuttle OR bilateral player motion****: end the rally at the last frame with competitive evidence from ***either***, and ****hard-terminate on a "shuttle picked up" event****. Two cheap detectors, no new model required:

55 - ***Player kinematics*** ? person velocity drops / players go stationary after a point (the casual returner taps it while the other walks). ****Reuse:**** the person cue producer (torchvision + ByteTrack) in ``yolo_hybrid`` (fusion P1, ``[verified]`` exists).

56 - ***Shuttle-in-hand / out-of-play*** ? ``shuttle_player_colocation`` (shuttle box ? person box = held) ``[verified]`` built in ``fusion_features.py`` (default-OFF, unmeasured on golden) from the fusion plan §4, ****or**** the no-new-model proxy: ****sustained WASB `Visibility=0`**** after the last fast stroke (held shuttle goes dark). Use it as a ****termination anchor + reverse-timeline search**** to the true end (last net-crossing / last fast-shuttle / last bilateral motion) ? the owner's idea, refined: the hold event is ***late***, so anchor-then-backtrack, don't use its timestamp as the end.

57 - ****Lever B ? recall**** (targets ****G1****) ``[design]``. Both systems miss real rallies ? detector sensitivity: proposal-windowing / velocity-threshold tuning, and the ****complementary-fusion**** angle below. Measure recall floor per video; the highest-leverage labels are the ***missed* rallies** (active learning, RALLY_QUALITY_RESEARCH §4.3).

58 - ****Lever C ? signal/ensemble fusion**** (targets ****G1****) ``[design]``. Local & Gemini (and shuttle vs person cues) hit ***different* rallies** ? a union/ensemble lifts recall. Already-built scaffolding: ``rally_gate.py`` ****Gate A**** (net-crossings, structural rally-state, ``[verified]`` built, eval-only ? wire to prod = cheapest precision win for G2) and the FusionSegmenter feature path.

59 - ****Pure-precision cleanup**** (****G2****): ``rally_gate`` Gate A kills single-feed / held-shuttle ***starts* structurally** (?0?1 net crossings vs a real rally's several) ? the cheapest G2 win and independent of the person stack.

60

9. assistant (2026-06-19T17:18:22.279Z)

Both are clearly live/tracked. Let me read the next docs: RALLY_DETECTION_QUALITY_REPORT, PROMINENT_COURT_DETECTION, and the others.

10. user (2026-06-19T17:18:23.704Z)

```
1      # Local, Cheap, AI Rally Detection for Racket Sports ? Quality Report (Checkpoint)
2
3      **Internal · honest measured results only · checkpoint 2026-06-17 · supersedes the
2026-06-14 draft**
4      **Project: badminton-highlight-indexer · toward a paper**
5
6      > *An honest empirical study of a shuttle-trajectory rally segmenter, its over-
segmentation fix, a
7      > task-faithful evaluation metric, a disciplined small-N promotion gate, and a first
real-footage
8      > validation against human ground truth and a cloud VLM.*
9
10     ---
11
12     ## Abstract
13
14     We present a fully local, low-cost rally-detection system for racket sports (badminton
first) that
15     segments a match video into rally `[start, end]` intervals. The system is built on a
**frozen
16     shuttle-trajectory substrate** (a WASB CNN that emits per-frame shuttle position), from
which a
17     velocity signal seeds recall-oriented candidate windows that a tiny pure-numpy logistic-
regression
18     scorer grades. Every additional cue enters as a confidence-weighted **feature column**
the scorer
19     learns to weight ? never a hand-coded branch ? and ships **default-OFF**, promoted only
on a measured
20     held-out lift. We evaluate against an owned golden corpus, now **n=11 labelled matches**
(grown from
21     n=6 this cycle, including **5 real owner-hand-labelled ground-truth clips**), treating
```

the labelling

22 vendor as an interchangeable oracle: the contract is with the golden labels, not their provenance.

23

24 Since the last checkpoint (2026-06-14) the headline result moved from "one promoted-then-reverted cue"

25 to a ****statistically significant over-segmentation fix****. Bounded windowing plus a rally-END

26 inactivity trim (the "cap=6 + R4" milestone) lifts our ****task-faithful tolerance-match F1** from 0.091

27 to 0.323** on n=11 (? +0.232, 95% CI [+0.153, +0.304], paired sign-test ****10/0, p=0.002****), drives the

28 ****merge-rate** from 0.694 to 0.150** (rallies-swallowed-by-a-merge, down on ****all 11**** videos), and

29 tightens the mean rally-END error from ****52.1 s** to **3.8 s****. A first ground-truth comparison against a

30 cloud VLM (Gemini) shows local reaching ****~86?88 %** of Gemini on rally detection** (F1@0.25) with the

31 rally-****START** already at parity** ? the entire residual gap is the rally-END. Negative and null

32 results are reported in full (person and audio fusion cues remain non-significant at n=6; a

33 net-crossing gate that wins offline does ****not**** transfer to the served operating point). We also

34 introduce a deliberately honest ****task-faithful evaluation metric**** (asymmetric tolerance-match with

35 strict 1:1 assignment) after finding that plain temporal-IoU is the wrong ruler for this task in both

36 directions. The contribution is less any single number than a ****method for being honest at small**

37 **scale**** ? and the path to a publishable result is now a short one: more labelled matches.

38

39 ---

40

41 ## Headline results at a glance

42

43 | | |

44 |---|---|

45 | ****Over-segmentation milestone**** (n=11, R2 tolerance-F1) | ****0.091 ? 0.323**** . ? +0.232

· CI [+0.153, +0.304] · sign-test ****10/0, p=0.002**** |

11. user (2026-06-19T17:18:24.023Z)

1 # Prominent-Court Detection ? kill background-court false positives (multicourt)

2

3 > ****Status:**** `DRAFT` ? design + experiment ****owner-gated** for any default flip** .

****Owner:**** Avi . ****Created:**** 2026-06-19 . ****Last updated:**** 2026-06-19

4 > ****Lifecycle:**** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when DONE)

5 > ****Tracking anchors:**** §8 progress tracker is the source of truth; pointer in memory `current-state`.

6 > ****Relation to existing docs:**** attacks gap ****G2**** ("precision ? local over-fires, worst on multicourt") from [`RALLY\_DETECTION\_GAP\_CLOSURE.md`](RALLY_DETECTION_GAP_CLOSURE.md); it is the ****shuttle-side**** counterpart of the person-side ****Gate B / court mask**** in [`MULTI\_SIGNAL\_FUSION\_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md). Complements the held-shuttle ****shuttle_player_colocation**** (G3) ? court-localization (G2) and held-shuttle (G3) are independent precision levers.

7 > ****Honesty note:**** claims are tagged `verified` (checked vs code/measured) / `design` (proposed). Every signal lands ****flag-gated, default-OFF****, promoted only on measured lift per [`EXPERIMENT\_HARNESS.md`](EXPERIMENT_HARNESS.md) + [[decision-rigor-norm]].

8

9 ---

10

11 ## 0. TL;DR

12 In **multicourt** footage the camera sees several courts. The shuttle detector (WASB) tracks *any* shuttle in frame, so a rally on a **background / adjacent court** becomes a predicted rally on the **main match** ? a false positive. The fix: identify the **prominent court** (the foreground court that occupies the majority of the frame, where the match being filmed is played) and **keep only rallies whose shuttle lives inside it**. Mechanically this re-defines the shuttle signal from `(x, y, visible)` to `(x, y, visible-AND-inside-the-prominent-court)` **before** windowing ? so a background-court shuttle reads as "not there" and never forms a rally. A flat 2D floor polygon is the MVP; a **pseudo-3D play-volume** (court floor + the air column above it) is the refinement so a high clear over the prominent court isn't wrongly rejected. Lands as a **default-OFF experiment**; the flip is gated on a Launch Report ([[launch-report-convention]]).

13

14 ## 1. Problem & motivating example `[verified]`

15 - **Measured gap (G2):** local over-fires ~2x vs Gemini, and the over-fire is **worst** on multicourt ? even among long (>7s) rallies, multicourt over-fires **1.58x** (background-court games are *real* long rallies, just on the wrong court). See `RALLY_DETECTION_GAP_CLOSURE.md` §3` + the P7 long-rally lens.

16 - **Concrete anchor** ? GX010142 rally #1 (2026-06-19). **Owner observation:** "the first rally is just a shuttle flying in the non-prominent court? while the prominent-court players have not started." Confirmed from the cached WASB trajectory (`output/wasb/GX010142_1080p__wasb.csv``): rally #1 (`0.72?3.65s``, 2.94s) has a shuttle centroid at **meanX ? 1616** on a 1920-wide frame (?84% to the right) ? spatially apart from the central cluster where most rallies live (meanX ? 1000?1300). Several other suspicious rallies (#3, #9, #12, #16) cluster in the same far-right band ? consistent with a **right-side adjacent court**.

17 - **Why a 1D threshold is not enough** `[verified]`: the courts do not separate cleanly on X or Y alone (rally Y-centroids overlap 250?620 px across courts; the far-right band mixes with foreground play). The courts are **2D regions** under perspective ? so we need the court *geometry*, not a coordinate cutoff. This is the owner's "pseudo-3D / court model" intuition, validated by the data.

18

19 ## 2. Goal & non-goals

20 - **Goal:** on multicourt clips, drop rallies whose shuttle is outside the prominent court, **without losing real prominent-court rallies** ? measured per-video, no regression on single-court clips (where it must be a no-op or a do-no-harm).

21 - **Non-goals (v1):** multi-camera; tracking players across courts; full court-line homography for scoring; anything biometric. Single-court footage is explicitly a **no-op** (one court ? everything is "prominent").

22

23 ## 3. Decisions locked

24 | # | Decision | Rationale |

25 |---|-----|-----|

26 | D1 | The prominent court is a **2D region (polygon)** in image space, refined to a **pseudo-3D play volume** | 1D thresholds don't separate courts under perspective (§1) |

27 | D2 | Gate the **shuttle trajectory** by the region **before windowing** (re-define visibility) | Stops a background rally at the source ? no window ever forms there. Reuses `point_in_polygon`` |

28 | D3 | **Default-OFF**, flag-gated; single-court = no-op | Zero-regression rollout ([[weak-signals-retained]]); no behaviour change unless a polygon is supplied + the flag set |

29 | D4 | Reuse the existing **court_polygon`** config + `point_in_polygon``; do NOT invent a new geometry stack | `FusionConfig.court_polygon`` + `fusion_features.point_in_polygon`` already exist `[verified]` |

30 | D5 | The prominent court can be **supplied (config)** OR **auto-derived**; auto-derivation is its own measured step | Config polygon is the cheapest first experiment; auto is the product path |

31

32 ## 4. Design

33

34 ### 4.1 The signal change (the core experiment)

35 Today the windower consumes the raw WASB trajectory: a per-frame `(Frame, Visibility, X, Y)``. The change is a **pre-windowing filter**:

36

37 ```

38 for each frame f: if shuttle_visible(f) and NOT inside_prominent_court(X_f, Y_f): mark f as NOT visible

```

39     ...
40
41     A background-court shuttle is then indistinguishable from "no shuttle" ? the velocity
windower never opens a rally there. This is purely **subtractive** on the shuttle stream (it can
only remove visibility, never add it), so on a correctly-drawn prominent court it cannot
manufacture rallies ? only suppress out-of-court ones. (Mirror of `rally_gate`'s "do-no-harm"
discipline.)
42
43     ### 4.2 Identifying the prominent court (four routes, cheapest first)
44     1. **Configured polygon** (MVP, `[design]?buildable now`): a normalized `[[x,y],?]`
court polygon per camera/clip, exactly the existing `FusionConfig.court_polygon`. Draw it once
from the dense shuttle/player cluster. Good enough to validate the whole concept on GX010142 +
the multicourt golden clips.
45     2. **Player-location auto-derivation** (`[design]`): run
`PersonDetector.detections_per_frame` `[verified]` exists`, take the **largest, most-central,
most-persistent** player cluster (the foreground players are bigger boxes lower in frame), and
fit the prominent-court polygon to where *those* players' feet live. "Prominent = where the
match's players are." This is why the owner asked to **extract person features** ? players
anchor the court.
46     3. **Shuttle-density auto-derivation** (`[design]`): the prominent court is the densest
sustained shuttle-activity region; cluster the visible trajectory and take the dominant mode.
Person-free fallback.
47     4. **Court-line detection** (`[design]`, heaviest): Hough/segmentation court lines ? the
largest court quad. Most accurate, most work; deferred.
48
49     v1 ships route 1 (config) + measures; route 2 (player-anchored) is the product upgrade.
50
51     ### 4.3 Pseudo-3D play volume (the refinement)
52     A flat floor polygon rejects a shuttle at the top of a high clear *over the prominent
court* (it projects above the floor quad). The pseudo-3D model treats the prominent court as a
**floor quad + a vertical air column**: accept a shuttle if its image point lies within the
floor polygon **expanded upward** by a height envelope (the max plausible shuttle height
projected through the camera). Practically: the floor quad plus an upward `pad` (perspective-
aware ? larger near the net/far baseline). Start with a generous fixed upward pad (cheap, MVP);
upgrade to a per-camera height envelope only if measurement shows real rallies being clipped.
This keeps the floor tight (rejects adjacent-court floor play) while not clipping legitimate
high shuttles.
53
54     ### 4.4 Reuse map `[verified]`
55     | Need | Existing code |

```

12. assistant (2026-06-19T17:18:27.557Z)

Now let me read the progress trackers / status sections of the remaining docs and the report's status footer. Let me check QUALITY_ITERATIONS, R2_EVAL_METRIC_DESIGN, MULTI_SIGNAL_FUSION_PLAN headers, and the progress trackers (§7/§8) of the two tracked docs and the report's conclusion.

13. user (2026-06-19T17:18:28.517Z)

```

1     # Rally-detection quality iterations ? living log
2
3     **Purpose.** A running, reverse-chronological log of how we evolve rally-detection
quality: each
4     iteration's hypothesis, what shipped, the measured (or expected) effect, and the lesson.
This is the
5     **live** continuation of the historical trail in
6     `[archives/quality-iterations/`](archives/quality-iterations/README.md) ? per the
archive policy
7     (`CLAUDE.md`), **terminal-state** (DONE/REJECTED) work gets a dated subdir there; this
file stays at
8     `docs/` top level and keeps growing, then its entries graduate to the archive when they
conclude.
9
10    **Methodology (non-negotiable).** Every quality change rides the experiment harness
11    ([EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md)): new cues land **flag-gated, default

```

```

OFF**; they're
12     graded **offline against golden labels** (per-video + LOVO-honest); promotion to default
happens **only
13     on measured lift** through the `run_regression.py` gate; **rollback = flip a config
flag, never
14     `git revert`.** Cues are designed in
[MULTI_SIGNAL_FUSION_PLAN.md](MULTI_SIGNAL_FUSION_PLAN.md); scoring
15     in [EVALUATION.md](EVALUATION.md); the 2-layer model in
16     [HOW_RALLY_DETECTION_WORKS.md](HOW_RALLY_DETECTION_WORKS.md).
17
18     **Golden corpus = the 15 labelled matches** in `output/human_lovo_manifest.json` (the
original 6 ?
19     `adarsh_avi_singles, kushagra_singles, largetest_doubles, gbaaddy, testlarge_short,
20     mahadevpura_singles` ? owner-confirmed 2026-06-13, since grown to 15 with the later
GoPro/Boxhill clips). The model trained on these is what we run alongside the existing
21     detector on NEW footage for the side-by-side review.
22
23     ---
24
25     ## Baselines that drive strategy (IoU 0.5 vs human golden labels)
26
27     | Path | Mahadevpura (clean-ish) | Kushagra (multi-court) |
28     |---|---|---|
29     | Heuristic (velocity windowing) | 0.657 | 0.157 |
30     | Gen-0 owned head (LOVO, n=6) | 0.744 | 0.561 |
31     | Two-stage WASB+Gemini refine | 0.83 | ? |
32     | Gemini wrapper (`gemini-flash-latest`) | 0.93 | 0.66 |
33
34     **Reading:** clean footage is commoditized (serve via Gemini); **multi-court drops the
commodity wrapper
35     to 0.66 ? that is the owned model's ship target.** Baselines tracked in
`eval_baselines/`. The owner's

```

14. user (2026-06-19T17:18:29.453Z)

```

1     # R2 ? Task-faithful rally-boundary eval metric (tolerance-match + over-seg guard)
2
3     **Status:** **IMPLEMENTED (augmenting), 2026-06-17** ? owner-reviewed + decisions locked
2026-06-17;
4     `backend/eval/tolerance_metrics.py` + `rally_seg_eval` reporting landed (default-
additive; tIoU stays the
5     gate pending the ablation experiment). Gates the rally-end work (R4) and the dual-eval-
set regression gate.
6     A focused increment of the rally-quality project
([RALLY_QUALITY_RESEARCH.md](RALLY_QUALITY_RESEARCH.md)) §7;
7     the day-1 validation that motivates it:
[REAL_FOOTAGE_VALIDATION_REPORT.md](REAL_FOOTAGE_VALIDATION_REPORT.md);
8     the directive: `eval-boundary-tolerance` + `eval-dual-set-regression`.
9
10    > **Implementation refinement (2026-06-17):** §2.3.2's boundary-error distribution is
computed over
11    > **unconditional best-overlap** matches, NOT the within-? 1:1 matches ? a ?-gated set
is survivorship-biased
12    > (loose ends fail the gate and vanish, hiding the very end-deficit the diagnostic
exists to surface; that
13    > within-? deficit instead shows up as low `tol_recall`). Matches the day-1 validation
method. Measured on the
14    > n=9 corpus: **|?start| 1.25 s vs |?end| 2.19 s** ? "start?solved, end=the gap",
confirmed.
15
16    ---
17
18    ## 0. TL;DR
19    `tIoU-F1@0.5` is the wrong ruler for rally detection, in **both** directions: it over-
penalizes the

```


20 **intrinsically fuzzy rally START** (the ~1?1.5 s service window), and its overlap
 matching is murky about

21 **over-production**. R2 replaces the headline with a **tolerance-match metric under
 strict 1:1 (Hungarian)

22 assignment**, reported as a **decomposition** ? `P@? / R@? / F1@?` + **boundary-error
 distributions split

23 into start/end** + an **over-segmentation guard** + a **?-sweep** ? not a single number.

24

25 **Honest, slightly uncomfortable consequence (measured, see §3):** under R2 the local
 detector's headline

26 number *drops* and the gap to Gemini *widens* vs what tIoU showed ? because R2
 faithfully charges for the

27 1.5× over-production and the loose ends that tIoU's overlap matching partly absorbed.
 That is the point:

28 R2 splits "behind Gemini" into its two real causes ? ** (a) over-production [precision]**
 and ** (b) loose

29 rally-ends [the R4 lever]** ? and forgives only the inherent start fuzziness. It is a
 more honest ruler,

30 not a flattering one.

31

32 ---

33

34 ## 1. Why `tIoU@0.5` is the wrong ruler (grounded in the n=2 ground-truth clips)

35 1. **Over-penalizes the START.** For a 5.7 s rally, `tIoU ? 0.5` tolerates only ~±1.9 s
 of boundary shift

36 (`? = D·(1??)/(1+?)`). The service window (stance ? racquet contact) alone is ~1?1.5
 s ? even **Gemini's**

37 median |?start| is 0.98?1.46 s. So tIoU spends most of its budget on a boundary that
 is *intrinsically*

38 undefined to ~1 s. (Owner directive: don't strictly align starts; tolerate ~±2 s.)

39 2. **Count parity is a deceptive proxy.** mahadevpura-2 hit 1.03× of Gemini's count yet
 F1@0.5 = 0.150 ?

40 right count, wrong boundaries. A count- or overlap-based view hides this.

41 3. **It flips config rankings on noise.** The hard-cap was golden-neutral at n=6 (+0.006
 n.s.) but "best" at

42 n=7 once a continuously-active clip entered ? a metric artifact at the 0.5 cliff, not
 a quality change.

43 4. **The deficits we must see are start?end-symmetric.** Local's |?start| ? Gemini's;
 the **entire** gap is

44 the rally-**END** (|?end| 2.1?3.1 s vs Gemini ~0.8?1.2 s). A single blended IoU
 cannot show that ? and

45 directing R4 needs exactly that decomposition.

15. user (2026-06-19T17:18:30.141Z)

1 # Multi-signal rally fusion ? add a human (person) cue alongside the shuttle

2

3 **Status:** **P1 DONE** (2026-06-13) ? the person cue is extracted to

4 `backend/pipeline/detectors/person\_detector.py` (`PersonDetector`), consumed by
 `yolo\_hybrid` with no

5 behavior change. **P2?P4 remain owner-gated.** Phases below are sequenced but each is a
 separate PR.

6 **Owner decisions captured** in [§9](#9-open-decisions-for-the-owner).

7

8 > **Companion docs:** the 2-layer mental model is
 HOW_RALLY_DETECTION_WORKS.md;

9 > the modular-fusion architecture this builds on is **ADR-007**
 (archives/decisions/DECISIONS.md);

10 > the permissive-dependency rule is **ADR-002 / ADR-005** ; the audio cue (the #1 next
 cue, already

11 > scaffolded) is [archives/research/AUDIO_RALLY_DETECTION_PLAN.md](archives/research/AUD
 IO_RALLY_DETECTION_PLAN.md).

12 > The vendor labels that gate the learned path come from
 GOLDEN_SET_VENDORING.md

13 > and the labeling spec in ROORA_LABELING_GUIDELINES.md.

```

14      > **How each cue here is added/tested without production regression** ? A/B, shadow
eval, and the
15      > promote-only-on-lift discipline ? is [EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md)
(every P1?P4 cue
16      > below lands flag-gated, default OFF, promoted only on measured LOVO lift).
17
18      ---
19
20      ## 1. Context ? why this, why now
21
22      Rally detection today runs **one cue at a time**. Each segmenter independently turns a
single signal
23      into rally windows:
24
25      | Segmenter | Signal | File |
26      |---|---|---|
27      | `wasb_hybrid` / `tracknet_hybrid` | shuttle velocity (frozen WASB/TrackNet trajectory)
| `backend/pipeline/detectors/tracknet_runner.py` |
28      | `yolo_hybrid` (the Pydantic `default_segmenter` fallback; `config.json` overrides it
to `gemini`) | person velocity (torchvision + ByteTrack) |
`backend/pipeline/segmenters/yolo_hybrid.py` |
29      | `motion` | raw pixel motion | `backend/pipeline/segmenters/motion.py` |
30      | `gemini` | a cloud VLM watching the whole clip | `backend/providers/gemini.py` |
31
32      **There is no fusion.** The owner's directive: make the detector **use the shuttle *and*
the humans
33      together** ? and be ready to fold in more signals (audio, later pose) ? delivered
34      **MIT / Apache-2.0 / BSD-clean** so we can ramp commercially without a licensing
rethink.
35
36      This is **not greenfield** ? it operationalizes work the repo already blessed:
37
38      - **`HOW_RALLY_DETECTION_WORKS.md` open question:** "Fusion: combine shuttle-motion
(WASB) with
39      player-motion for static cams."
40      - **ADR-007** already designed a **modular "rally layer"** that fuses cues *outside* the
frozen WASB
41      model, with **audio as the #1 next cue and person/pose parked as #2**. This plan picks
up the parked
42      person cue and generalizes the fusion seam so future cues are "register a producer,"
not "re-touch
43      the segmenter."
44      - **`backend/pipeline/detectors/rally_gate.py` already exists** ? the net-crossing gate,
explicitly
45      labelled in its own docstring as "Gate A in the over-fire fix spectrum (**B = player-
stance state

```

16. assistant (2026-06-19T17:18:34.088Z)

Now let me check the RALLY_DETECTION_QUALITY_REPORT's status/conclusion footer and the progress trackers (\$7 of RALLY_QUALITY_RESEARCH, \$7 of GAP_CLOSURE, \$8 of PROMINENT_COURT). Also need to grep for inbound references and check archives for existing copies.

17. user (2026-06-19T17:18:36.657Z)

```

quality-iterations
AUDIO_RALLY_DETECTION_PLAN.md
RALLY_DETECTION_RESEARCH.md

```

18. user (2026-06-19T17:18:36.746Z)

```

docs\archives\COMMERCIAL_READINESS_REVIEW.md
docs\archives\MONETIZATION_AUDIT.md
docs\archives\checkpoints\2026-06-eval-baseline\README.md

```

docs\archives\checkpoints\2026-06-logging-reporting\README.md
docs\archives\checkpoints\2026-06-scaffolding-complete\README.md
docs\archives\past_projects\README.md
docs\archives\research\AUDIO_E1_FINDINGS.md
docs\archives\research\AUDIO_RALLY_DETECTION_PLAN.md
docs\archives\research\RALLY_DETECTION_RESEARCH.md
docs\archives\checkpoints\2026-06-strategy-foundation\README.md
docs\archives\quality-iterations\2026-06-human-gt-tuning\README.md
docs\archives\quality-iterations\2026-06-initial\README.md
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md
docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md
docs\archives\decisions\DECISIONS.md
docs\archives\TEST_RUN_SUMMARY.md
docs\archives\quality-iterations\2026-06-first-ab-experiment\README.md
docs\archives\checkpoints\2026-06-14-hardening-loop-complete.md
docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md
docs\archives\DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md
docs\archives\HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md
docs\archives\README.md
docs\archives\quality-iterations\2026-06-18-lovo-resweep-n13\README.md
docs\archives\quality-iterations\README.md

19. assistant (2026-06-19T17:18:39.962Z)

None of my 7 batch docs have an already-archived copy with the same stem (the archived `RALLY\_DETECTION\_RESEARCH.md` is a different doc from `RALLY\_DETECTION\_GAP\_CLOSURE.md`/`RALLY\_QUALITY\_RESEARCH.md`). Now let me grep for inbound references to each filename across the repo, and check the status footers of the report + progress trackers.

20. user (2026-06-19T17:18:42.019Z)

CLAUDE.md:2
docs\ANALYZERS_AND_RECONCILERS.md:6
docs\CODE_MAP.md:1
docs\DESKTOP_APP_PLAN.md:1
backend\config\models.py:3
docs\GOLDEN_DATA_SHARING.md:1
eval\baselines\fixtures\README.md:1
docs\EXPERIMENT_HARNESS.md:4
artifacts\pr245_review.md:2
docs\HOW_RALLY_DETECTION_WORKS.md:1
docs\archives\CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md:1
artifacts\pr245_review.json:2
artifacts\pr245_fixes_plan.md:1
docs\GOLDEN_VIDEOS.md:1
backend\pipeline\detectors\person_detector.py:1
backend\pipeline\detectors\court_gate.py:1
docs\GOLDEN_REGRESSION_FIXTURES.md:3
backend\pipeline\detectors\tracknet_runner.py:1
docs\PERSONALIZATION_PLAN.md:1
docs\NEXT_STEPS.md:3
docs\RALLY_DETECTION_GAP_CLOSURE.md:7
docs\R2_EVAL_METRIC_DESIGN.md:4
docs\VIDEO_RECORDING_GUIDELINES.md:2
docs\RALLY_QUALITY_RESEARCH.md:12
docs\REAL_FOOTAGE_VALIDATION_REPORT.md:2
docs\R1_MILESTONE_LAUNCH_REPORT.md:1
docs\README.md:7
backend\eval\windowing.py:1
docs\ROORA_LABELING_GUIDELINES.md:2
docs\QUALITY_ITERATIONS.md:5
backend\eval\tolerance_metrics.py:1
docs\PROMINENT_COURT_DETECTION.md:4

```

tests\test_config.py:1
docs\PROJECT_DOC_TEMPLATE.md:1
docs\POST_166_WORK_ITEMS_MILESTONES_AND_RISK_ASSESSMENT.md:1
tests\test_court_gate.py:1
docs\archives\quality-iterations\2026-06-18-lovo-resweep-n13\README.md:3
backend\eval\rally_seg_eval.py:3
scripts\nightly_regression.py:4
backend\eval\experiment.py:1
backend\eval\golden_fixtures.py:1
docs\archives\quality-iterations\2026-06-first-ab-experiment\README.md:1

```

Found 102 total occurrences across 42 files.

21. assistant (2026-06-19T17:18:45.412Z)

High inbound reference counts across code and docs. These are deeply wired-in live docs. Let me confirm the per-filename "cited as live" status by checking NEXT_STEPS and CLAUDE.md specifically, and verify the report's status footer plus the progress trackers to confirm none claim terminal state.

22. user (2026-06-19T17:18:47.489Z)

```

17:  `docs/HOW_RALLY_DETECTION_WORKS.md` (2-layer mental model) ?
`docs/MULTI_SIGNAL_FUSION_PLAN.md`
60:-  **Substantial design/project work gets a tracked project doc** following
`docs/PROJECT_DOC_TEMPLATE.md` (status header · decisions-locked · progress tracker ·
definition-of-done; archived on completion). Precedents: `docs/RALLY_QUALITY_RESEARCH.md` $7,
`docs/GOLDEN_REGRESSION_FIXTURES.md`.

```

23. user (2026-06-19T17:18:49.174Z)

```

59:-  **Docs:** `docs/MULTI_SIGNAL_FUSION_PLAN.md` (shuttle + person + audio, one fusion layer) ·
70:-  **External due-diligence review inputs (2026-06-17) ? PICK UP AFTER current ideas are
exhausted; full details + citations in [`RALLY_QUALITY_RESEARCH.md`](RALLY_QUALITY_RESEARCH.md)
$5.6. ** An owner-commissioned 31-reference review of the quality report *validated* the
methodology and surfaced these **future** levers (research inputs, not active work, not new
claims; verify citations before any submission):
96:   (`RALLY_QUALITY_RESEARCH.md` $5.6) ? better fit for the cheap/local/fast mandate.

```

24. assistant (2026-06-19T17:18:52.211Z)

Both CLAUDE.md and NEXT_STEPS.md cite these as live/authoritative. Now let me verify the report's conclusion/status (does it claim COMPLETED?) and the progress trackers of the tracked docs to confirm none are terminal.

25. user (2026-06-19T17:18:55.542Z)

```

=== QUALITY_REPORT tail ===
-  **Lead with the tolerance-match metric + over-segmentation metrics** (merge-rate, F1@k), not
bare tIoU.
-  **Per-cue ablation with CIs** on the grown corpus (the `ablation.py` harness already exists).

**Target venues** (applications/systems-oriented): **CVIU, ICIP, IEEE Trans. Multimedia, or a
CVPR/
ICCV workshop** (CV-for-sports / fine-grained video). Not the CVPR/ICCV main track (no
algorithmic
novelty). The cheapest path to a publishable result remains **more labelled matches**, not more
engineering.

---

```

10. Engineering & reproducibility

- ****Experiment leaderboard**** ? every ``compare()`` appends one reproducible row to

```

`eval_baselines/experiment_leaderboard.json`, pinned to `(spec_hash, label_set_hash, IoU)`, so
any
  regression traces to a recipe change, not drift.
- Nightly regression tripwire ? CPU-only (cached trajectories), guards DEFAULT (served) and
MILESTONE
  configs; exit 1 = regression, 4 = stale baseline.
- Standing litmus ? `served_gate_a.py` pins the honest eval?serve facts.
- CI gates ? Ruff, lenient-but-green mypy with a ratchet, syntax+import smoke (no GPU
stack), and the
  full offline mocked suite with a coverage floor. Combined with default-OFF cues + a pinned
CONTROL
  variant, rollback is a config flip, never a `git revert`.
- Privacy-safe fixtures ? golden-regression fixtures + de-attribution tooling allow video-
free
  shipping of the eval sets.

---

```

11. Conclusion

The contribution of this work is less any single F1 number and more a **method for being honest at small scale** ? and this cycle shows that discipline paying off. On an owned golden corpus grown to **n=11** (now with real ground truth), bounding the windowing and trimming the rally-END produced the project's **first statistically significant rally-quality win** (tolerance-F1 0.091 ? 0.323, p=0.002), drove the merge-rate down on **every** video (0.694 ? 0.150), and ? measured against human ground truth for the first time ? put the rally-**START** at parity with a cloud VLM, localizing the **entire** remaining gap to the rally-END. We reported the nulls in full (person, audio), banked the eval?serve lesson (the reverted Gate-A ? the honest W2), and introduced the task-faithful metric that made the over-production tIoU hid finally visible. The architecture stays deliberately standard where the literature is mature; we claim novelty only on the task-faithful metric and the small-N promotion discipline. The path to a publishable result is now short and concrete: **more labelled matches, multi-court isolated, ablations on the grown corpus** ? the moat was always the consented data plus the honest harness, not the weights.

```

=== GAP_CLOSURE $7 tracker ===
4:> Lifecycle: `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when DONE)
5:> Tracking anchors: $7 progress tracker is the source of truth; pointer in memory
  `current-state`; add to `docs/README.md` + `docs/NEXT_STEPS.md` when work starts.
65:- Nothing here is shipped or even started ? implementation is owner-gated. A DONE row in
$7 means *measured on golden*, not *flipped on by default*.
69:## 7. Deliverables & progress tracker ? source of truth
70:Legend: ? Todo ? In progress ? Done ? Gated. One small PR per row, off `origin/master`,
no stacking.
74:| P0 | Golden scoring loop (`ingest_golden.py` + `boxhill17_score_golden.py`); GX010141
ingested (#12) + scored | ? | No | ? | this work |
75:| P1 | Ingest the rest of the Boxhill golden batch (137/020140 + a large-count clip) ? per-
video baseline table | owner labels | ? owner | ? | ? |
76:| P2 | Wire `rally_gate` Gate A (net-crossings) to the served path ? cheapest G2 win | ?
| ? owner | ? | ? |
77:| P3 | Extract person-velocity + `shuttle_player_colocation` on the golden clips (default-OFF
features) | P1 | ? owner | ? | ? |
78:| P4 | Lever A: player-state rally-END rule (R-next) ? measure ?F1 / ?merge-rate /
?\\?end\\ vs R4 | P3 | ? owner | ? | ? |

```

79:| P5 | **Lever B/C**: recall + complementary fusion ? measure recall lift, no precision regression | P3 | ? owner | ? | ? |
80:| P6 | Launch decision: milestone + best levers default-ON, with a Launch Report (both rulers + over-seg guard) | P4,P5 | ? owner | ? | ? |
81:| P7 | **Long-rally (>5s) quality lens** (eval `--min-duration`/`--stratify`) + reel knob (`stitching.min\_rally\_duration`), both default-OFF. **Measured n=15**: single-court over-fire is short-FP-dominated (seg_ratio 1.43?1.04) but **multicourt over-fires 1.58x** even on long rallies (? court-localization, Lever C) and strict tolP doesn't rise (over-merge-contaminated). See [`QUALITY\_ITERATIONS.md`](QUALITY_ITERATIONS.md) "P7". | ? | No (eval+cfg, default-OFF) | ? measured | this PR |
88:## 9. Definition of done
93:- [] ? flip `Status ? DONE`, archive to `docs/archives/`.

26. assistant (2026-06-19T17:19:03.502Z)

This confirms everything. The GAP_CLOSURE doc explicitly states "Nothing here is shipped or even started ? implementation is owner-gated" and its §7 tracker has mostly ?? rows ? clearly LIVE in-progress work. The QUALITY_REPORT is a dated checkpoint with future-facing next steps (no COMPLETED/terminal header). All 7 docs are LIVE.

Let me do a final confirmation on the report's status header and whether any doc is a duplicate of an archived copy ? none have matching archived stems. The report header says "checkpoint 2026-06-17 · supersedes the 2026-06-14 draft" ? it supersedes an earlier draft but is itself the current checkpoint, not superseded. All evidence gathered.

27. user (2026-06-19T17:19:40.187Z)

Structured output provided successfully

28. assistant (2026-06-19T17:19:51.311Z)

All 7 docs in my batch are KEEP_LIVE. Summary of findings:

- **RALLY_QUALITY_RESEARCH.md** ? ? Foundation/tracked-project doc, definition-of-done not yet met, cited as authoritative in CLAUDE.md + NEXT_STEPS.md (12 inbound refs). LIVE.
- **RALLY_DETECTION_GAP_CLOSURE.md** ? DRAFT execution tracker; §6 explicitly says "Nothing here is shipped or even started," §7 rows mostly ?? owner-gated. LIVE in-progress.
- **RALLY_DETECTION_QUALITY_REPORT.md** ? current 2026-06-17 paper checkpoint that *supersedes* the older draft (it is not itself superseded); forward-looking §9/§11 + active paper aim. LIVE.
- **PROMINENT_COURT_DETECTION.md** ? brand-new (2026-06-19) DRAFT design/experiment, owner-gated, actively being built (`court\_gate.py` exists). LIVE.
- **QUALITY_ITERATIONS.md** ? self-described living log that "stays at docs/ top level and keeps growing"; only concluded entries graduate to dated archive subdirs. LIVE by design.
- **R2_EVAL_METRIC_DESIGN.md** ? "IMPLEMENTED (augmenting)" but not terminal ("tIoU stays the gate pending the ablation experiment"; gates open R4 work); backs live `tolerance\_metrics.py`. LIVE.
- **MULTI_SIGNAL_FUSION_PLAN.md** ? only P1 DONE, P2?P4 owner-gated; cited live in CLAUDE.md + NEXT_STEPS.md. LIVE.

No already-archived duplicates exist for any of these stems (the archived `RALLY\_DETECTION\_RESEARCH.md` is a distinct doc). No DEDUPE_STUB or ARCHIVE_NOW cases, and no contradiction traps requiring ASK_OWNER ? none of the 7 claim a terminal COMPLETED/DONE/REJECTED state while being cited live. Nothing to archive from this batch.